

Billy Marcus Wright

Marcus03wright@icloud.com | (830) 554-0614

www.linkedin.com/in/marcus-wright-523830297/ | www.themarcuswright.com

EDUCATION

Swarthmore College | Swarthmore, PA

May 2025

B.A. in Computer Science with Minor in Philosophy; *Cumulative GPA: 3.88/4.0*

- Sigma Xi Associate Member; Questbridge Scholar
- Relevant Coursework: Machine Learning, Deep Learning (Directed Reading), Operating Systems, Software Engineering, Data Structures and Algorithms, Open Source Software Development, Linear Algebra, Calculus 1 & 2

Cranfills Gap High School | Cranfills Gap, Tx

May 2021

Valedictorian; *Cumulative GPA: 4.0/4.0*

- Senior Class President; Leadership in sports/organizations such as Tennis, Basketball, Track, One-Act Play, 4H, FFA.

TECHNICAL SKILLS

- Proficiency in Python, and experience with C, C++, JavaScript, and Java
- Experience with Machine Learning (scikit-learn, pytorch), Human-Computer Interaction (HCI), Linux, FOSSoftware, ReactJS, MongoDB, GitHub, HTML

PUBLICATIONS

- Albatool Wazzan, Marcus Wright, Stephen MacNeil, and R. Souvenir. 2025. Evaluating the Impact of AI-Generated Visual Explanations on Decision-Making for Image Matching. In Proceedings of the 30th International Conference on Intelligent User Interfaces (IUI '25). Association for Computing Machinery, New York, NY, USA, 672–684.

PROFESSIONAL EXPERIENCE

Full-Stack Developer | TraffickCam (Non-Profit) | Part-Time

August 2024 – Present

- Manage front- and back-end production and dev servers. Maintain and update features/services across full stack.
- Performed a complete overhaul of Search functionality to improve model index modularity and scalability. Updated Inpainting Model, improving inpainting performance by more than 20% while prioritizing CPU computational cost.
- Responsible for various production-level UI updates and fixes across primary interface. Primary back-end engineer, responsible for bug fixes, updates, and maintenance for dev and production instances.
- Experience with Flask, SQL, ReactJS, CSS, CI/CD, GitHub, FAISS, Reverse Image Search, Linux, Pandas, Node
- TraffickCam info can be found at <https://traffickcam.com/investigative-platform/> and <https://traffickcam.com/>

PROJECTS & RELEVANT EXPERIENCE

National Science Foundation (NSF) Research | Temple University

June 2024 – July 2024

- Conducted research for the NSF REU Site for Pervasive Computing at Temple University, within the Vision, Imaging, and Data Analysis Research (VIDAR) Lab under Dr. Richard Souvenir's mentorship. Resulted in publication above.
- Focused on evaluating AI explanations, analyzing their impact on improving human interaction and decision-making with AI systems. Gained experience with Human-Computer Interaction, and Explainable AI (xAI) systems.
- Integrated AI explanation methods into a Deep Similarity ML Model and added to a web-based frontend to facilitate participant interaction with the AI explanations. Conducted a user study with modified frontend.

Project for Machine Learning | Swarthmore College

October 2024 – December 2024

- Designed and implemented advanced imputation strategies and various other feature engineering methods to optimize model inputs. Tested for the most optimal combination of methods to improve model accuracy.
- Evaluated multiple preprocessing pipelines across KNN, Logistic Regression, Decision Trees, and Random Forests, using statistical validation (ANOVA, Kruskal-Wallis) to assess their effect on predictive accuracy (~84-85%)

Project for Software Engineering | Swarthmore College

October 2023 – December 2023

- Built a client-server architecture on a web and Android mobile application using Node Express to handle http requests and responses with MongoDB as the database. I worked with 2 other students on this 7-week project using agile methodology. Application was designed to manage funding and inventory for student organizations.

LEADERSHIP & ADDITIONAL EXPERIENCE

Advisory Council to the CS Department (ACCS) | Swarthmore College February 2024 – May 2025

Student Board Member [Aug 2024 – Present]

- Assisted and designed various programs and events to encourage activity and inclusivity within the department.

Mentorship Program Mentor [Feb 2024 – May 2024] [Sep 2024 – Present]

- Mentored under-classmen majoring in Computer Science at Swarthmore College.

Teaching Assistant for Networks (CS43) | Swarthmore College September 2024 – December 2024

- Held weekly office hours to help students with homework assignments and labs.

Co-Ed Club Volleyball | Swarthmore College August 2022 – May 2025

Senior Advisor [Aug 2024 – Present]; Co-President [Aug 2023 – May 2024]; Treasurer [Aug 2022 – May 2023]

- Oversee activities of the Club Volleyball Executive Board and work closely with the Student Budgeting Committee and Office of Student Engagement to manage the club. Managed Club funding. Trained other officers.
- Organize and lead weekly practices. Work closely with students of various backgrounds, ethnicities, and religions.

Swarthmore College Libraries, McCabe A&US | Swarthmore College September 2023 – May 2025

Special Library Assistant [Sept 2024 – Present]; Library Assistant [Sept 2023 – May 2024]

- Assist students and patrons in accessing and using McCabe Library resources.

AWARDS

NSF CISE REU Travel Grant March 2025

- Received an NSF CISE REU Travel Grant to attend Intelligent User Interfaces in March 2025.

Questbridge National College Match Scholarship Recipient and Questbridge Scholar December 2020

- Received a full-ride Scholarship through the Questbridge National College Match to attend Swarthmore College.

DAR Good Citizen Award May 2021

- Honored for exemplifying qualities of dependability, service, and leadership within school and community.